

**INDIAN INSTITUTE OF
INFORMATION TECHNOLOGY
GUWAHATI**



**AI LAB
Assignment-7**

Name: Ansik Singh Tomar
Roll No: 2401037
Section: CS21
Branch: B.Tech 2nd Year CSE
Course: Artificial Intelligence Lab
Code: CS236

Travelling Salesman Problem using Hill Climbing and Simulated Annealing

1. Sample Inputs and Outputs

Input 1

Number of Cities: 5
Cost Matrix:

0 10 15 20 25
10 0 35 25 17
15 35 0 30 28
20 25 30 0 23
25 17 28 23 0

Output

Hill Climbing
Path: 0 -> 2 -> 3 -> 4 -> 1 -> 0
Cost: 95
Simulated Annealing
Config 1 Path: 0 -> 1 -> 4 -> 3 -> 2 -> 0
Config 1 Cost: 95
Config 2 Path: 0 -> 2 -> 3 -> 4 -> 1 -> 0
Config 2 Cost: 95
Config 3 Path: 0 -> 2 -> 3 -> 4 -> 1 -> 0
Config 3 Cost: 95

Input 2

Number of Cities: 7
Cost Matrix:

0 12 10 19 8 14 16
12 0 3 7 2 6 20
10 3 0 6 20 4 9
19 7 6 0 4 12 8
8 2 20 4 0 11 5
14 6 4 12 11 0 13
16 20 9 8 5 13 0

Output

Hill Climbing
Path: 0 -> 2 -> 5 -> 1 -> 3 -> 6 -> 4 -> 0
Cost: 48
Simulated Annealing
Config 1 Path:0 -> 4 -> 6 -> 3 -> 1 -> 5 -> 2 -> 0
Config 1 Cost: 48
Config 2 Path: 0 -> 1 -> 4 -> 6 -> 3 -> 2 -> 5 -> 0
Config 2 Cost: 51
Config 3 Path: 0 -> 4 -> 6 -> 3 -> 1 -> 5 -> 2 -> 0
Config 3 Cost: 48

2. Simulated Annealing Parameter Comparison

N	Configuration	Temp(T)	Cooling (α)	Iteration (K)	Tour Cost	Observation
5	Config 1	1000	0.995	2500	95	Balanced exploration and convergence, good quality solution
5	Config 2	500	0.99	1250	95	Faster convergence but sometimes trapped in local minima
5	Config 3	2000	0.9	5000	95	High exploration but unstable convergence, longer runtime
7	Config 1	1000	0.995	4900	48	Good balance between exploration and convergence
7	Config 2	500	0.99	2450	51	Faster but more likely stuck in local minima
7	Config 3	2000	0.9	9800	48	High exploration but slower and unstable convergence